PEBBL user manual

Physiology in the Emotion, Brain & Behavior Lab

Please Eyeball Before Believing Labels.

For research assistants. Updated 8 November 2026. If anything here doesn't match what you see, tell the lab staff.

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1. What PEBBL does

Our studies record several body signals at once, such as heartbeat, finger pulse, breathing, sweat, blood pressure and face-muscle activity. A computer finds the heartbeats and flags stretches that look unusable, but it makes mistakes. PEBBL shows you each recording with the computer's marks on it, and you check them by eye and fix them. Two research assistants check each recording separately, and a reconciler then settles any disagreements.

PEBBL never changes the recordings. It saves your marks in a separate file next to the recording in the data folder (for our lab, on Box), named with your initials.

The work comes in three steps, which you pick in PEBBL's form:

- **Step 1: Build ECG QRS template.** Once per participant for each of you (every RA builds their own template), before you review their ECG. You correct 20 seconds of heartbeats, and PEBBL uses them to find heartbeats in the rest of the recording.
- **Step 2: Review a channel.** Most of your work: one signal (a "channel") of one run at a time.
- **Step 3: Reconcile two reviewers.** For reconcilers only.

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2. Set up your computer (once)

On a lab computer, PEBBL is already installed. Skip to step 6 (make your own working folder), then start PEBBL with the **PEBBL** icon on the desktop. Sign in to Box Drive with your own account if it asks.

On your own computer, allow about 30 minutes, mostly waiting. You need an internet connection and permission to install programs. Wherever these steps say `YOU`, use your own username on that computer.

1 Install Python 3.11

Python is the programming language PEBBL is written in. You won't need to write any code.

WINDOWS

Open python.org/downloads/release/python-3119, scroll to *Files*, and download **Windows installer (64-bit)**. Run it. On its first screen, tick **Add python.exe to PATH**, then click **Install Now**.

MAC

Open the same page and download **macOS 64-bit universal2 installer**. Open it and click through the installer.

Use 3.11 even if a newer Python exists: PEBBL's parts are matched to it. Having other versions installed too is fine.

2 Install Git

Git downloads PEBBL and keeps it up to date.

WINDOWS

Download Git from git-scm.com/download/win, run it, and accept every default (keep clicking **Next**, then **Install**).

MAC

Open **Terminal** (press `⌘` `Space`, type *Terminal*, press `Return`). Type `git --version` and press `Return`. If a window offers to install the command line developer tools, click **Install**.

3 Get access to the data

PEBBL reviews recordings kept in your study's **data folder**, organized as in section 10. Each study has its own. In our lab, data folders are on Box. If the computer doesn't have Box Drive yet, install it from box.com/resources/downloads and sign in with your Tufts account.

The lab staff tell you where your study's data folder is. Check that you can open it: it holds one folder per participant (`sub-001`, `sub-002`, …) and the study's settings file, `pebbl_study.toml`. If you can't open it, ask the lab staff for access.

4 Download PEBBL

WINDOWS

Open **PowerShell**: Start menu, type *PowerShell*, press `Enter`.

MAC

Open **Terminal**: press `⌘` `Space`, type *Terminal*, press `Return`.

Paste these two lines, pressing `Enter` (`Return` on a Mac) after each:

```
cd ~
git clone https://github.com/tufts-ebbl/pebbl.git
```

This makes a folder called `pebbl` in your home folder. **Leave it there.** Don't move it into OneDrive, iCloud, Box or a synced Desktop or Documents folder: they would try to sync PEBBL's 1 GB of parts.

5 Run the one-time setup

WINDOWS

In File Explorer, open `C:\Users\YOU\pebbl` and double-click **Set up PEBBL.bat**.

MAC

In Terminal, paste this and press `Return`:

```
bash ~/pebbl/setup_pebbl.sh
```

A window shows progress for 5 to 15 minutes and ends with *PEBBL is set up*. If it ends with *Setup didn't finish*, take a photo of the window and send it to the lab staff.

6 Make a working folder

Make an empty folder called `PhysioWorking` in your home folder (`C:\Users\YOU\PhysioWorking`, or `/Users/YOU/PhysioWorking` on a Mac). While you work, PEBBL copies one participant's files here, and it deletes the copy once your work is safely back in the data folder. PEBBL keeps each study's copies in a folder of its own inside it (for example `PhysioWorking\urglab`), so there's nothing to set up if you work on more than one study. On a shared computer, everyone makes their own.

7 Start PEBBL

WINDOWS

Double-click **Start PEBBL.bat** in the `pebbl` folder. For a desktop shortcut, right-click it → **Send to** → **Desktop (create shortcut)**.

MAC

Double-click **Start PEBBL.command** in the `pebbl` folder (Finder → **Go** → **Home** → `pebbl`). To keep it handy, drag it to the right-hand end of your Dock.

Each time, PEBBL first checks for updates, then opens its form. A text window (the **terminal**) stays open behind the form, and PEBBL writes notes there during your session. After each session the form opens again; click **Cancel** on it to close PEBBL. **Don't close the terminal** until it says *PEBBL has closed*.

8 Try the demo

In the form, choose **Step 2**, then tick **Use synthetic demo data** (ticking it locks the Step choice), type your initials and click **Start**. You get made-up practice data with no real participant in it, so explore freely: nothing you do here is kept.

If setup goes wrong

YOU SEE	WHAT TO DO
Python 3.11 isn't installed	Install it as in step 1. On Windows, use the python.org installer, not the Microsoft Store version.
git is not recognized	Install Git (step 2), then close PowerShell or Terminal and open it again.
Couldn't update PEBBL	Usually no internet connection. PEBBL still starts with the version you have; tell the lab staff if it keeps happening.
Mac: Start PEBBL.command won't open	Run `bash ~/pebbl/setup_pebbl.sh` again; it makes the file runnable.
Anything else	Take a photo of the window and send it to the lab staff.

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3. Before your first real session

- **Get certified** on the practice participant first (section 7). PEBBL scores your practice against an answer key and shows you what to fix.
- **Use the same initials every time:** 2 to 4 letters, such as `jkl`. They name your files. The first time you use them, PEBBL asks you to confirm them. If it warns that `sub` looks like the start of a subject ID, check what you typed.
- **Know your assignment.** The lab staff assign participant, run and channel in the Google tracking spreadsheet. Set that channel to *in progress* when you start and *finished* when you finish. PEBBL reminds you at both points.
- **Protect the data.** These recordings come from real people. Don't screenshot them or share them outside the lab, and keep them in the working folder PEBBL manages.

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4. The signals in plain words

You mark two kinds of things. **Peaks** are single points, one per heartbeat or pulse. **Bad stretches** are spans of time where a signal can't be trusted. A stretch you leave unmarked in a finished review counts as checked and clean.

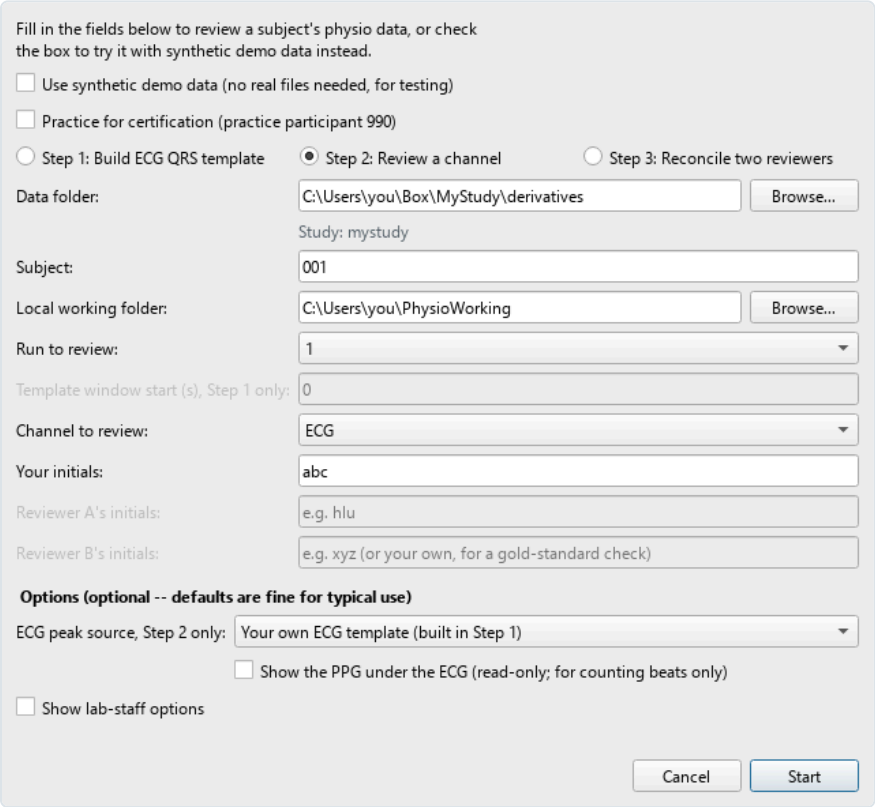
CHANNEL	WHAT IT IS	WHAT YOU MARK
ECG	The heart's electrical activity, from stickers on the chest. Each heartbeat makes one tall, sharp spike: the **R peak**.	A peak on every R peak

CHANNEL	WHAT IT IS	WHAT YOU MARK
PPG	The pulse in a fingertip, from a sensor on a finger. Each heartbeat makes one smooth wave, reaching the finger a fraction of a second after the R peak.	A peak on every pulse
RSP	Breathing, from a stretchy belt around the torso. One rise and fall per breath. Some studies show it as `rsp_raw`, the belt as recorded.	Bad stretches
EDA	Skin conductance: tiny changes in sweat on the hand. Smooth rises lasting a few seconds are real responses.	Bad stretches
COR	Activity of the frowning muscle above the eyebrow (corrugator EMG), in studies that record it. Shown twice: in fine detail, and at full range underneath.	Bad stretches
SBP, DBP	Blood pressure, in studies that measure it at every heartbeat with a finger cuff: the upper number (systolic) and the lower (diastolic), in mmHg.	Bad stretches
Event	Which part of the task was running: 0 is outside the task, other numbers are task phases.	Nothing (context only)

Some channels also show a **guide row** underneath: another signal shown for context, which you can't mark. Section 6 says how to use each one.

5. A review session, start to finish

Fill in the form

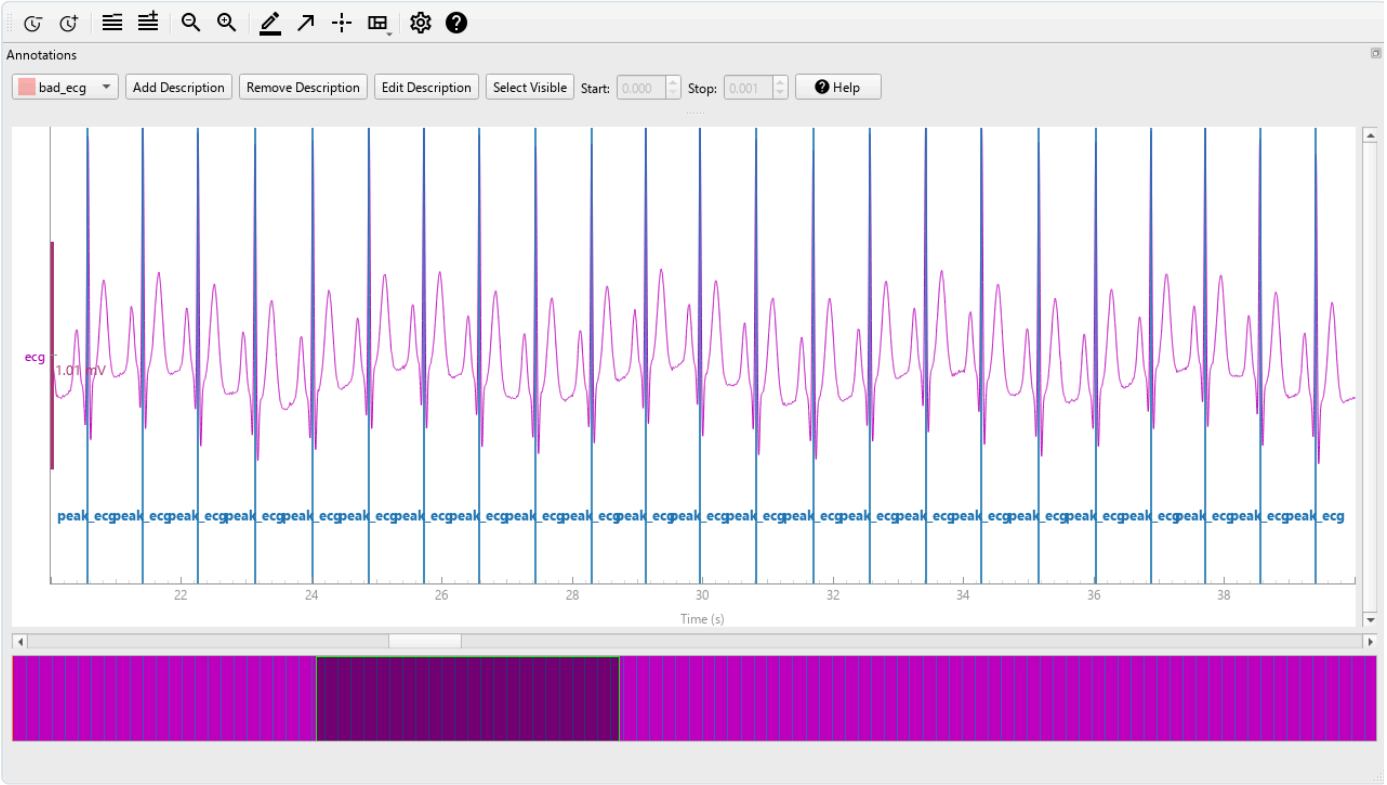


The form, ready for an ECG review of participant 001, run 1.

- **Step:** usually *Step 2: Review a channel*.
- **Data folder:** click **Browse** and choose your study's data folder from setup step 3, the one that holds the participant folders (`sub-001` , `sub-002` , …). Underneath, PEBBL shows the study it belongs to. If that isn't your study, or it says something is missing, choose again; if it's the right folder, ask the lab staff. PEBBL remembers the folder.
- **Subject**, **Run** and **Channel to review:** from your assignment. `1` , `001` and `sub-001` all work as the subject.
- **Local working folder:** your `PhysioWorking` folder.
- **Your initials.**
- **Options:** leave them as they start. Don't tick *Show lab-staff options*.

Click **Start**. PEBBL copies the participant's files to your working folder and opens the viewer. **Read the notes in the terminal window:** they say what's already marked and anything special about this recording.

Find your way around the viewer



An ECG review in annotation mode. Each blue line is a peak mark. The label list at the top left says which kind of mark a drag will make: here it's on `bad_ecg`, so switch it to `peak_ecg` before adding peaks.

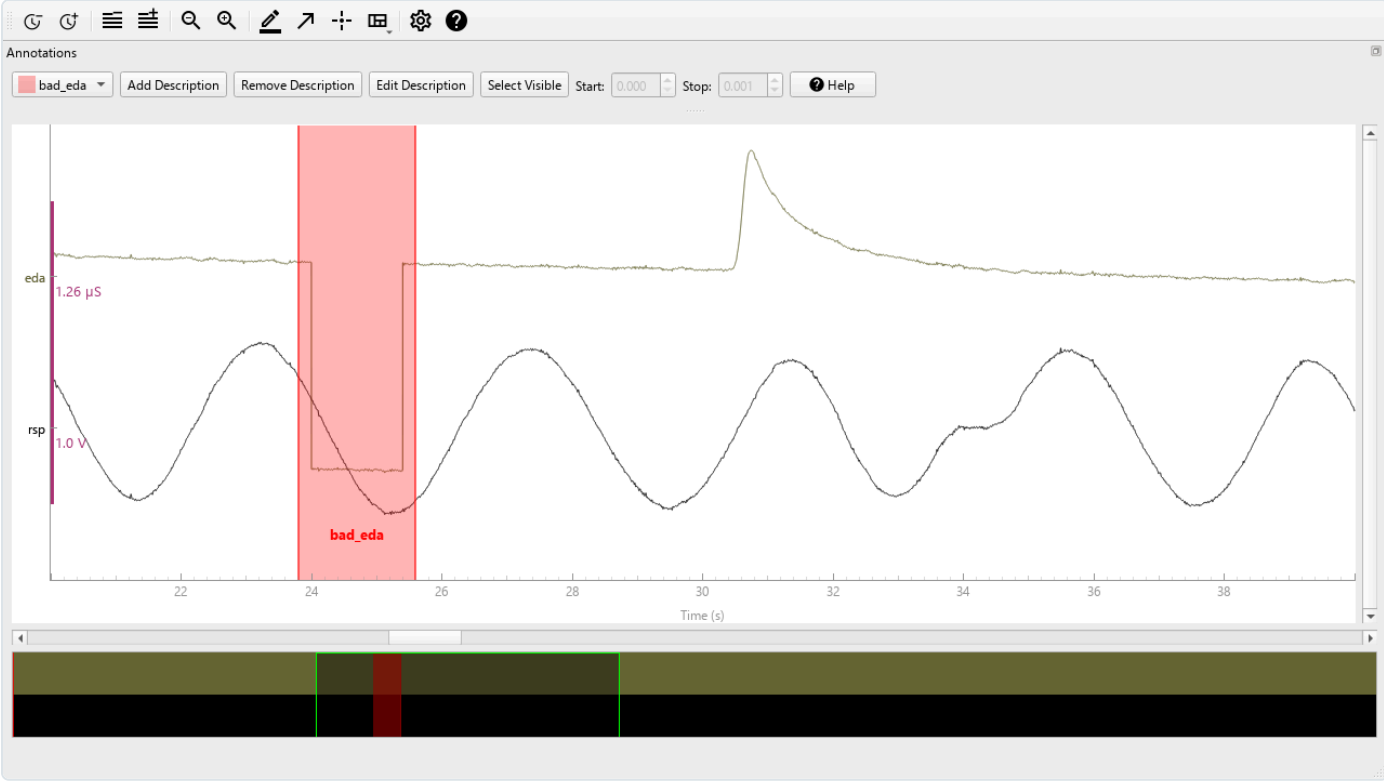
TO	DO THIS
Page through the run	`Shift` + `→` moves one full screen, `→` a quarter. Or drag the scroll bar, or click the overview strip at the very bottom.
Start marking	Press `a` for **annotation mode**. A bar appears with a label list.
Read an exact value	Press `x`, then hover. The value and time show at the bottom.
Hide the task markers	Press `e`.
See every shortcut	Click the **?** button in the toolbar.

Add, delete and move marks

Check the label before every drag. The label list often starts on a bad-stretch label such as `bad_ecg`. A drag made with the wrong label makes the wrong kind of mark.

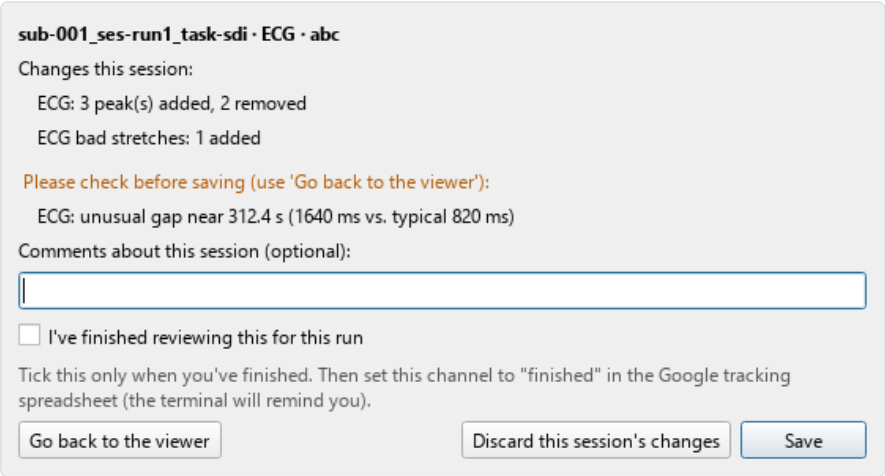
- **Add a peak:** choose the peak label (`peak_ecg` or `peak_ppg`) and drag a small box over the top of the wave. PEBBL saves the mark at the highest point inside your box.

- **Delete a peak:** with the peak label chosen, drag a small box over the mark (the two merge into one box), click the box, then right-click it.
- **Move a peak:** delete it, then add a new one. **Never drag a new box over a mark you want to keep:** the two merge, and one is lost.
- **Mark a bad stretch:** choose the channel's bad label (`bad_ecg`, `bad_ppg`, `bad_rsp`, `bad_eda`, `bad_sbp`, `bad_dbp` or `bad_emg_cor`) and drag across the stretch. Drag its edges to resize it; right-click it to delete it. If PEBBL says the computer's stretches on a channel count whatever you do, leave them and only add your own.



A bad stretch over an EDA artifact: the signal drops and snaps back, as when an electrode lifts. The smooth rise near 31 s is a real response, so it stays unmarked. The breathing (RSP) guide row is underneath.

Close the viewer and save



The session summary appears when you close the viewer.

- **Read what it asks you to check,** such as an unusual gap between beats. Click **Go back to the viewer** to look. Leaving a flagged spot as it is is fine if the signal really looks like that.
- **Comments:** note anything odd for the lab staff.
- **I've finished reviewing this for this run:** tick it only when you've checked the whole run, start to end, including the parts before and after the task.
- **Save.** *Discard this session's changes* throws them away and keeps your earlier saved work.

PEBBL then copies your file back to the data folder, checks it arrived, and deletes the local copy. Then the form opens again, ready for your next session. Update the tracking spreadsheet after each session. When you're done for the day, click **Cancel** on the form and wait for *PEBBL has closed* before closing the terminal window.

You can stop partway. Save without ticking *finished*. Next time, start PEBBL with the same participant, run, channel and initials, and it picks up where you left off.

6. Channel by channel

ECG Peaks on every R peak

Step 1 first, once per participant (each of you builds your own). PEBBL shows 20 seconds of ECG with the computer's R peaks. Fix them and close the window. A preview then shows your beats as gray lines with their average in black. Click **Approve Template** if the beats line up. If they're messy, click **Try a Different Window** and run Step 1 again with *Template window start* set to 20 (then 40, and so on).

FIX

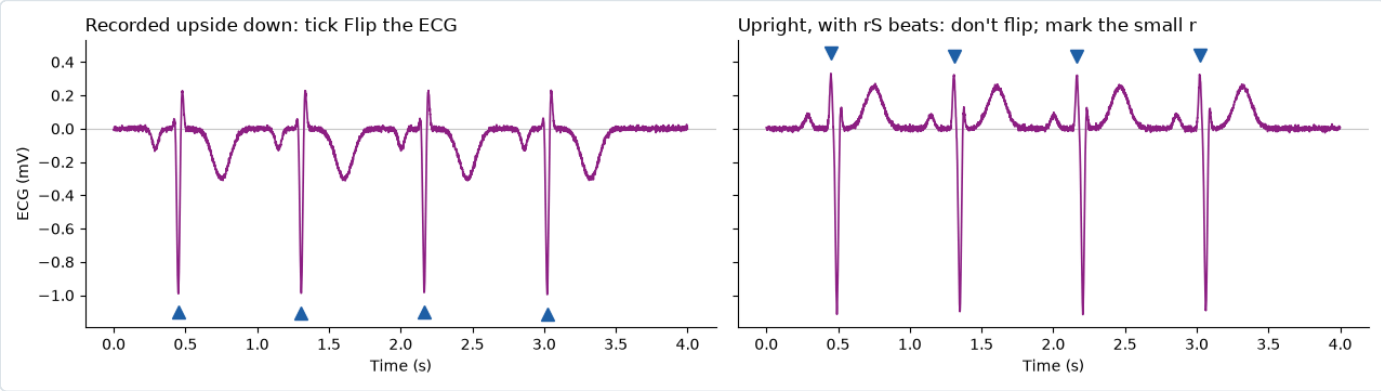
- Add a peak on any heartbeat without one.
- Delete peaks on T waves (the rounder bump after each beat) or on noise.
- Mark `bad_ecg` only where you can't tell where the R peaks are. Peaks inside a bad stretch are dropped automatically.

LEAVE ALONE

- Premature beats: an early, sometimes odd-looking beat is real.
- Natural changes in heart rate.

The finger-pulse (PPG) guide row is **off** under ECG. You can turn it on in the form to check your count: every pulse comes a fraction of a second after an R peak. If a pulse has no mark before it, look in the ECG for a missed R peak. Place each peak by the ECG itself, not by the pulse: the gap between them varies a little from beat to beat (and, in studies that record the pulse on a separate device, from run to run). An early beat may make only a small pulse or none, and it's still real.

Flip the ECG (only in studies whose form shows the box). In most recordings each beat's tall, sharp spike points up. If the spikes point **down** and the T waves (the rounder bump after each beat) point down with them, the ECG was recorded upside down: tick **Flip the ECG** in the form, for Step 1 and Step 2. Some upright recordings have **rS beats**: a small upward spike (the r) followed by a deep downward one (the S), with the T wave pointing **up**. Don't flip those, and mark the small r, not the S or the little bump just after it. The T waves tell the two apart.



Simulated ECGs. Left: recorded upside down, so the spikes and the T waves point down; tick *Flip the ECG*. Right: upright, with rS beats: a small upward r, then a deep downward S, and T waves pointing up; don't flip it, and mark the small r. The triangles show where each beat's mark belongs.

PPG Peaks on every pulse, in a session of their own

FIX

- **To add a peak, drag a box over the whole top of the pulse,** both humps if it has two. PEBBL puts the mark on the highest point.
- Delete peaks in dropouts: low, smooth arcs with no clear pulse.
- Move marks the summary lists as *not on a crest* (they're on a slope) onto a crest.
- Mark `bad_ppg` where you can't see the pulses, such as during a hand movement (big, irregular waves).

LEAVE ALONE

- A mark on either hump of a two-humped pulse. Don't move marks from one hump to the other.
- The small pulse right after an early heartbeat. It's a real pulse: give it a peak on its tallest bump, as for any other pulse.
- Never add a peak where you can't see a pulse, even if the rhythm says one should be there.

The computer's dropouts start marked as `bad_ppg`, and peaks inside them aren't loaded. Delete or shrink a marked stretch if the pulse there is clearly fine.

RSP Bad stretches only

MARK BAD

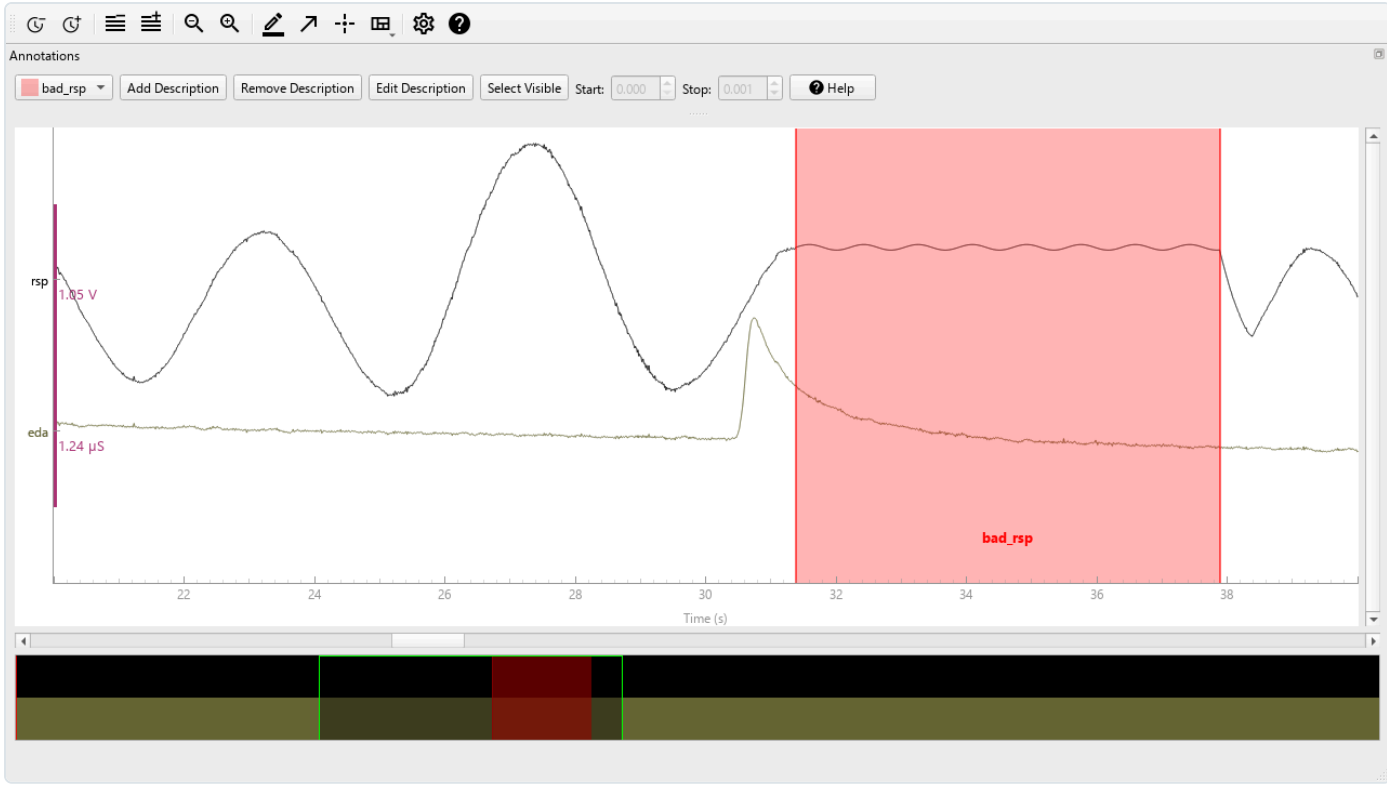
- **Anywhere you can't see where each breath peaks:** you can't place a peak you can't see.
- A flat line with no breaths, whether the belt stopped recording or a breath was held.
- Clipped tops: breaths cut off flat at the very top of the recording's range.
- A sudden drop or jump much bigger than a breath, often at a task event, then a slow drift back.

LEAVE ALONE

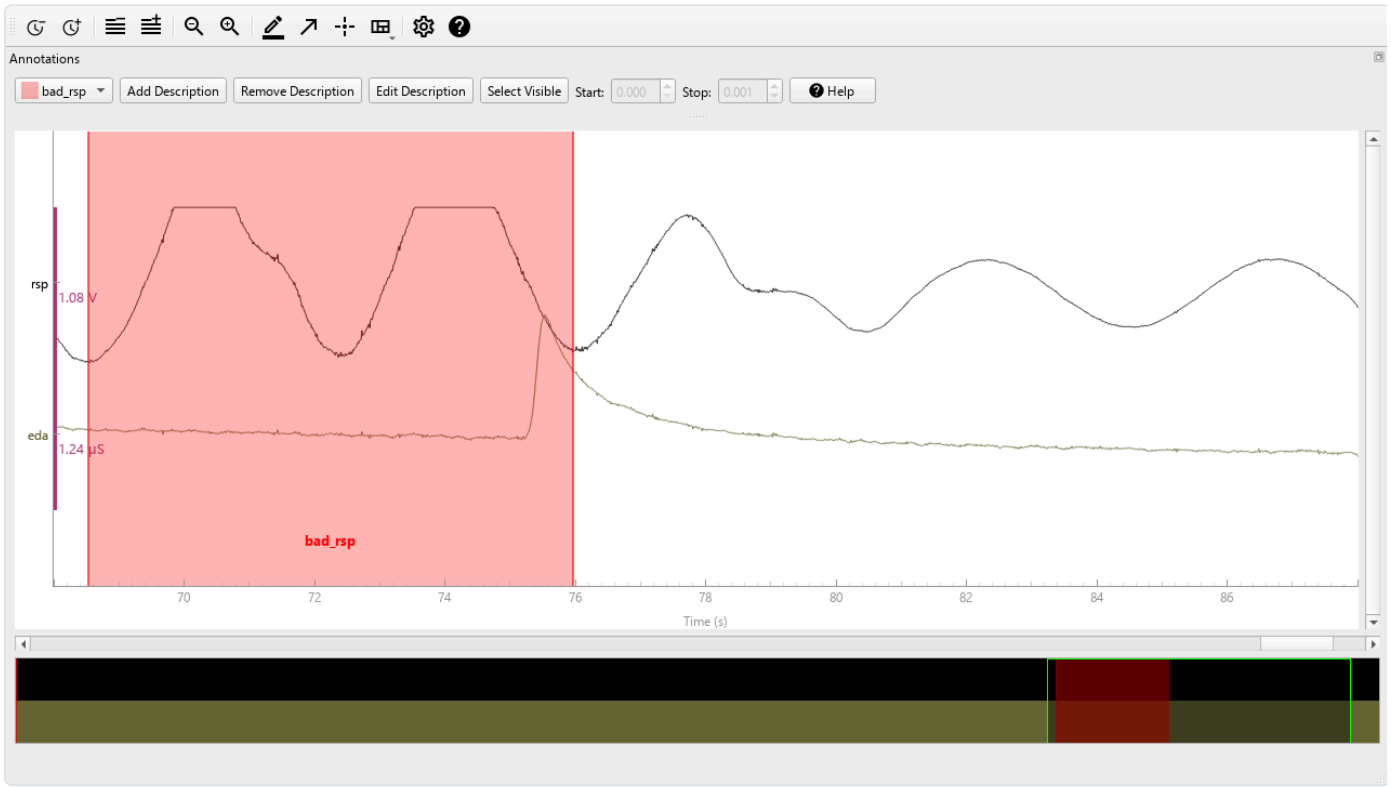
- **Breaths whose peaks you can see,** however odd: sighs, shallow breathing and pauses between breaths are all real.

The review starts with the computer's flags; check each one. The EDA guide row underneath helps: a rise in EDA 1 to 3 seconds after a big breath shows the breath was real. Never mark RSP bad because of EDA.

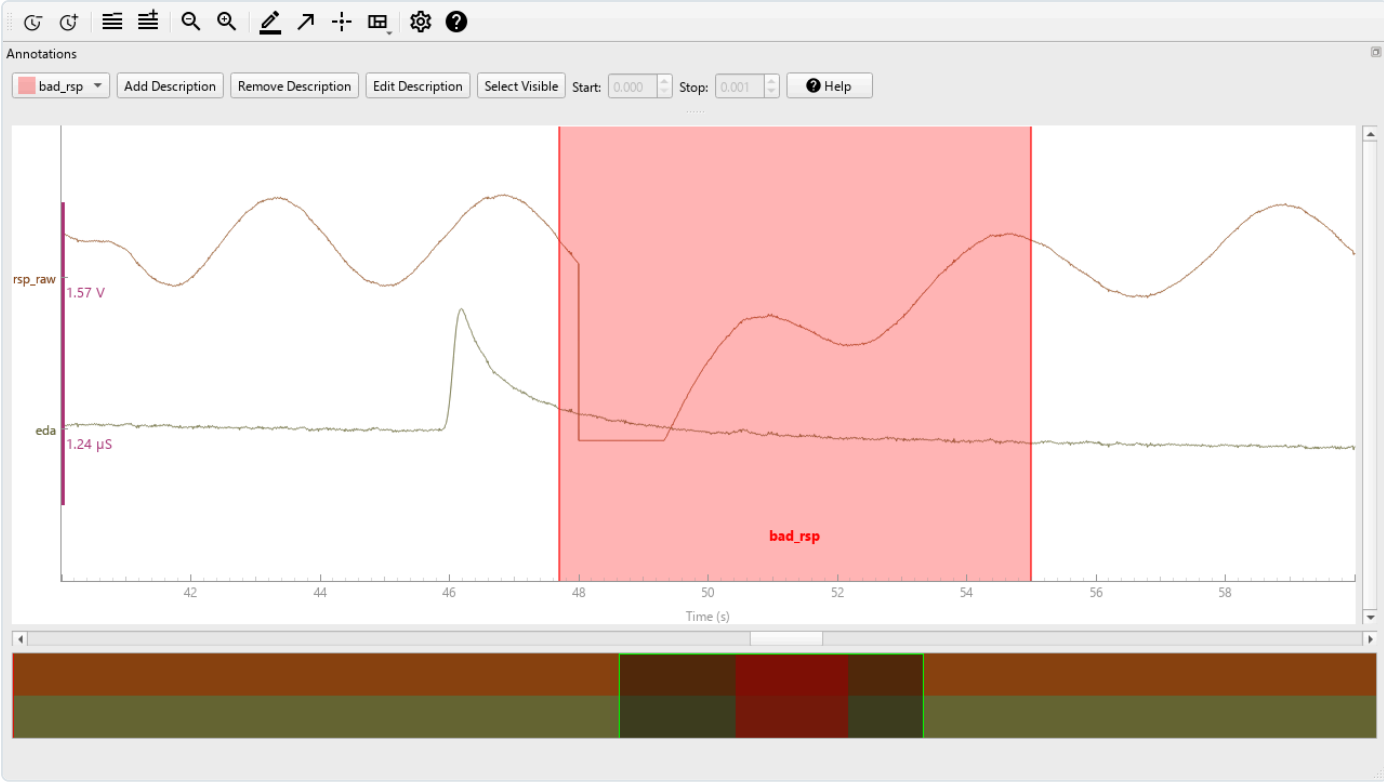
In some studies the RSP row is `rsp_raw` : the belt as recorded, before any filtering. Judge and mark there, with `bad_rsp` as usual; your marks are saved for RSP. Anything much taller than a breath is cut off flat a little beyond the row. A study may also leave some of the computer's flags off the screen, such as the first and last seconds of a recording. The opening lines in the terminal say which. Those count in the analysis whatever you mark.



A held breath, marked `bad_rsp`: the line goes flat, so there are no breath peaks to see. The deep breath before it is a sigh. It's real, so it stays unmarked, and the EDA rise after it confirms it.



Clipped tops, marked `bad_rsp`: two deep breaths cut flat at the top of the recording's range. The shallow breaths after them still show their peaks, so they stay unmarked.



A belt swing, marked `bad_rsp`, in a study that shows `rsp_raw`: a sudden drop much bigger than a breath, then a slow drift back. The mark runs from just before the drop to where ordinary breathing is back. The bottom of the drop is cut off flat because it's much taller than a breath.

EDA Bad stretches only

MARK BAD

- Sudden drops or jumps, as when an electrode lifts.
- Sharp spikes, and stretches that go flat.
- Brief artifacts: the computer misses anything shorter than about 5 seconds.

LEAVE ALONE

- Real responses: a smooth rise over 1 to 3 seconds, then a slow decline.

The breathing guide row underneath helps: a rise right after a big breath is a real response, not bad signal. Mark EDA bad only where the EDA itself is unusable.

SBP and DBP Bad stretches only

MARK BAD

- Sudden, implausible spikes.
- A value held flat for a long time (the cuff recalibrating).

LEAVE ALONE

- The computer's marks on blank stretches with no reading, including short ones under 2 seconds: they're gaps between readings.
- BP where the terminal says pulse samples were *lost in transfer*: the cuff still measured it.

The review starts with the computer's marks: no reading, values out of range, SBP less than 10 above DBP, and no usable finger pulse. The finger-pulse guide row underneath shows why the last kind matters: where there's no usable pulse, the pressure values aren't real.

COR Bad stretches only

Corrugator EMG, the frowning muscle above the eyebrow, in studies that record it. You mark `bad_emg_cor` stretches only. Your study's rules for what to mark are printed in the terminal when the session opens, under *About the channel(s), from the file*. Follow them.

COR is shown twice. `emg_cor` shows fine detail, such as the texture of quiet muscle next to a perfectly smooth dead line. Anything taller than its row is cut off at the row's edge, so a burst, a frown and a much bigger artifact all look equally tall there. `emg_cor_full`, right underneath, shows the same signal at full range: the tallest thing in the recording fills its row. Compare heights there. You can't mark on it separately: a stretch you mark covers both rows.

The form may list channels your study doesn't record. Choosing one just tells you there's no data to review.

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7. Get certified

Before you work on real data, you practice on **participant 990** (in some studies, on 991 too: the form lists your study's practice participants). It's a made-up recording: a computer simulated every channel, so no real person is in it. The computer's marks on it include deliberate mistakes, and a few spots look odd but are real. PEBBL keeps an **answer key** with it, the gold standard for where every peak and bad stretch belongs. The key is exact because the recording was built from it. Each time you save a practice review, PEBBL compares your marks with the key.

1 Tick *Practice for certification*

It's near the top of the form. PEBBL fills in subject 990 and uses the `practice` folder next to your data folder. If your study has more than one practice participant, a **Practice participant** list appears: practice on each one, choosing it there. Keep your usual data folder, working folder and initials. Step 3 isn't part of practice.

2 Build the ECG template (Step 1)

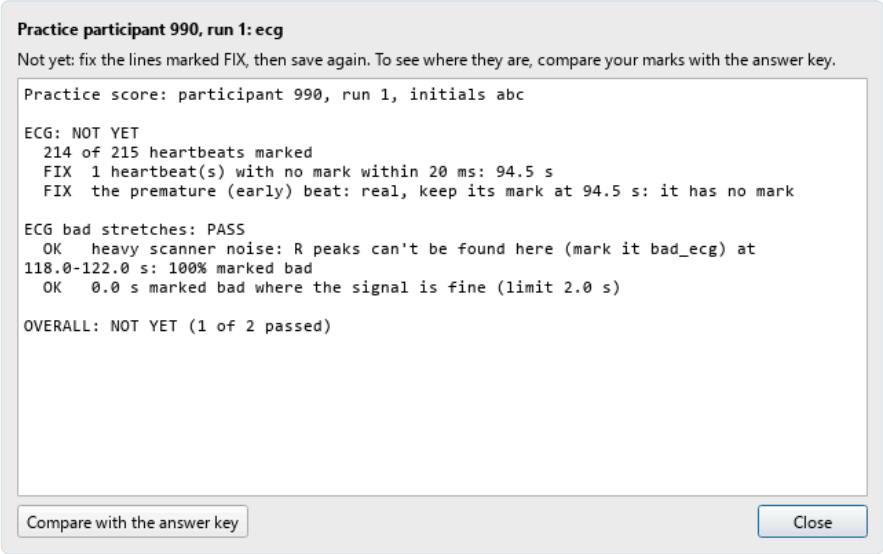
As in section 6. Step 1 isn't scored, but your Step 2 heartbeats start from it.

3 Review each assigned channel (Step 2)

The lab staff tell you which runs and channels to practice. Review as you would real data, and save.

4 Read your score

After you save, a score window opens. Each channel gets **PASS** or **NOT YET**. Each **FIX** line says what's wrong and where, in seconds from the start of the run.



This trainee deleted the mark on a premature beat at 94.5 s. The beat is real, so the mark should stay.

5 Compare with the answer key

Click **Compare with the answer key** to open the viewer with your marks and the key's together. Labels ending `_agree` are where you match the key. Labels ending `_only_key` are marks you missed, and labels ending in your initials are marks the key doesn't have. This view is only for looking: nothing you change in it is saved.

6 Fix and save again

Start Step 2 again for the same run and channel. PEBBL picks up from your last save. Fix the FIX lines, save, and repeat until the channel passes.

What counts as a pass

MARKS	TO PASS
Peaks (ECG, PPG)	Every heartbeat or pulse has a mark within 20 ms (ECG) or 30 ms (PPG) of the key's, and there are no extra marks. On a two-humped pulse, either hump counts. Beats inside a stretch you correctly marked bad don't need marks. Beats in a run's first half second and last second aren't scored, so a mark there never counts as extra.
Flip the ECG (studies whose form shows the box)	Ticked for an ECG recorded upside down, not ticked for an upright one. Set the wrong way, ECG doesn't pass, whatever the marks.
Bad stretches (all channels)	At least 80% of each real artifact is marked bad. Real signal that looks odd, and the computer's mistaken marks, are no more than 25% marked (a frown on COR: no more than 10%). In all, no more than 2 seconds are marked bad where the signal is fine. Up to 1 second on either side of a real artifact counts as part of it (2 seconds for a finger-pulse dropout). Stretches the computer marked that count whatever you do must be left as they are. A few stretches count either way, marked or not, such as the slow drift back after a breathing-belt step. A **MISS** line names something worth marking that doesn't stop a pass.

When every channel you practiced passes, tell the lab staff. Your score is also saved in your session log, so they can check it there.

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8. Step 3: reconciling two reviews

For reconcilers. Choose **Step 3**, the run and channel, your own initials, and the two reviewers' initials. Each mark shows with one of three labels: agreed by both (`_agree`), only the first reviewer, or only the second. Delete marks you don't believe, add anything both missed using the `_added` label, and leave agreed marks alone. Save as usual. For PPG, two marks on one pulse must be fixed before PEBBL lets you save.

9. When something goes wrong

- **Tell the lab staff:** what you were doing (participant, run, channel, step), plus a photo of the window.
- **PEBBL keeps a session log** next to your file in the data folder, ending in `_annotations_YOURINITIALS.log`. The lab staff can read what happened there.
- **Closed a window by accident?** Your last save is safe; unsaved changes are lost. Start again and PEBBL resumes from your last save.
- **Saved an ECG review with *Flip the ECG* set the wrong way?** Set the box right, redo Step 1, then open Step 2 for ECG: PEBBL offers to start the ECG over. Your earlier ECG marks are kept in your file, and your other channels stay as they are.
- **PEBBL says files from an earlier session were never copied back to the data folder:** answer `y` to copy them there.
- **PEBBL says there's a local copy from another study:** nothing has been changed. Do what the message says: start a session for that study's participant, or ask the lab staff.
- **A message says `processing_log.csv` is open:** close it in Excel. Your marks are already saved.
- **The form says the data folder has no study settings file, or isn't a data folder:** choose your study's data folder (setup step 3). If it's the right folder, ask the lab staff.

10. How the data folder is organized

In our lab, each study's processing pipeline already lays out its data folder this way, so RAs can skip this section. It's for lab staff, and for anyone setting up PEBBL for a study.

The data folder holds the study's settings file and one folder per participant. Each run of a recording sits in its own folder, with a JSON file of the same name beside it:

```
data folder/
  pebbl_study.toml
  sub-001/
    ses-run1/
      beh/
        sub-001_ses-run1_task-sdi_physio.tsv.gz
        sub-001_ses-run1_task-sdi_physio.json
    ses-run2/
      beh/
        sub-001_ses-run2_task-sdi_physio.tsv.gz
        sub-001_ses-run2_task-sdi_physio.json
  sub-002/
    …
```

- **The study's settings:** `pebbl_study.toml`, a short text file the lab staff write for each study: its name, its task name, and anything it does differently, such as which runs and channels it has. PEBBL won't review a data folder without one, and it records the study's name in every review. The developer README says how to write one.
- **Participants:** `sub-` plus the ID. IDs shorter than three characters get leading zeros, so participant 1 is `sub-001`.
- **Runs:** `ses-run1` and, if there is one, `ses-run2`. A study with one recording per participant only needs `ses-run1`.
- **File names:** `sub-ID_ses-runN_task-NAME_physio.tsv.gz`, where `NAME` is the study's task name, as in its settings file (for example `sdi`). Keep only one recording per run folder.
- **The recording** is a tab-separated table, compressed with gzip, with one row per sample and no header row.
- **The JSON file** gives the `SamplingFrequency` in Hz and the `Columns`, in order. PEBBL uses columns named `ecg`, `ppg`, `rsp`, `eda`, `emg_cor`, `emg_zyg`, `sbp`, `dbp`, `finger_temperature` and `event`. It also uses `ecg_peaks` and `ppg_peaks`: 1 where the computer marked a peak, 0 elsewhere. A recording can have only some of these columns.

PEBBL adds its own files beside each recording, named with the reviewer's initials. It keeps a `processing_log.csv` in the data folder. For certification, a `practice` folder sits next to the data folder.
